

Forkable sandboxes let computers explore executable alternatives.

Self-organization → self-driving cars
→ self-driving compute.

→ Reuse reached state. Run alternatives. Verify the result.

Yossi Eliaz

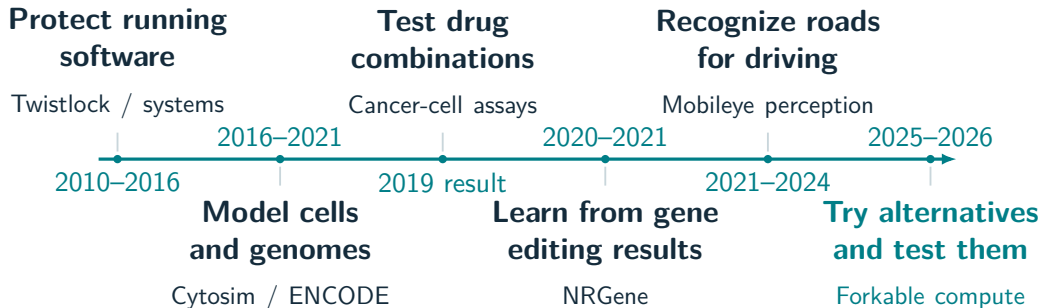
Principal Engineer, Incredibuild

Associate Professor, HIT

Cambridge Systems Research Group, 15 October 2026. Research and engineering perspectives; employer affiliation disclosed.

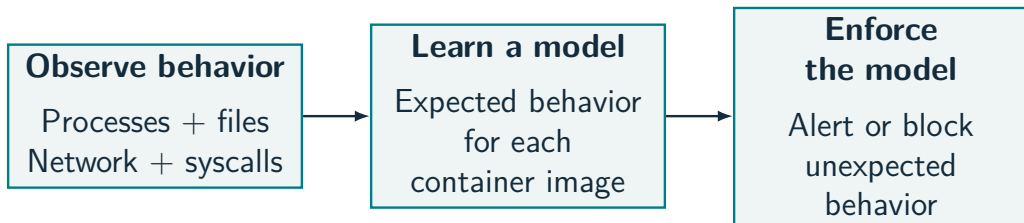
1 MOTIVATION

My work led from securing software and modeling cells to self-driving compute.



Career periods; cancer-drug result (2019). Work periods overlap.

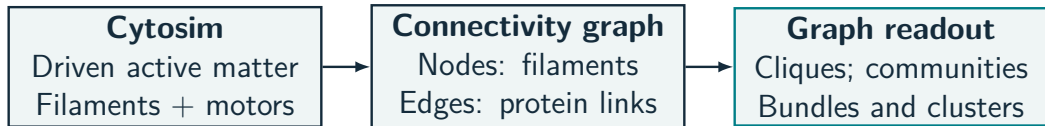
Twistlock learned container behavior and enforced the resulting model.



Vendor-reported learning plateau: usually about **1 hour** of cumulative runtime

Twistlock 1.7 Tech News, January 2017. Product/team mechanism.

Graph measures link protein binding to actomyosin self-organization.



2–7 binding sites per linker **600 s** simulated time **30** random starts

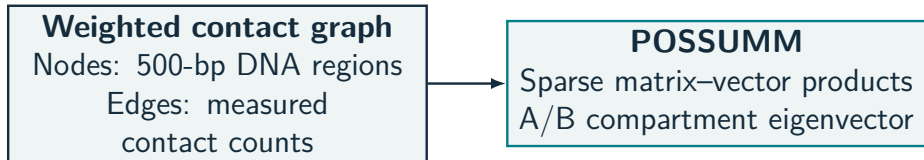
More binding sites promoted bundles and larger clusters.

Biological motivation: contraction and dendritic-spine plasticity involved in memory.

Eliaz, Nedelec, Morrison, Levine & Cheung, [Phys. Rev. E 102:062420 \(2020\)](#).

POSSUMM computed 500-bp chromosome compartments with 23 GB of RAM.

Hi-C is a weighted contact graph between DNA regions.



23 GB RAM
Measured

2.5 min
Measured

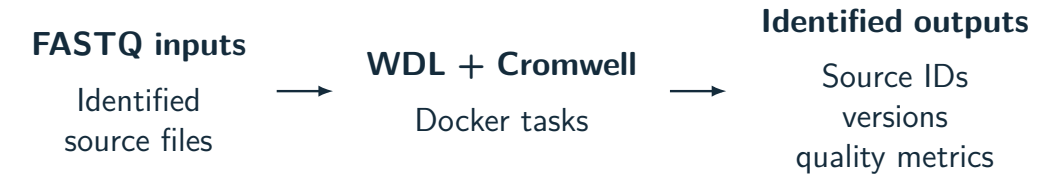
> 4.6 TB RAM
Dense projection

Chromosome 1, 500 bp. A/B identifies active and inactive chromatin.

Harris et al., [Nature Communications 14:3303 \(2023\)](#), Box 1 and Supplementary Fig. 2.

ENCODE linked reproducible execution to identifiable outputs.

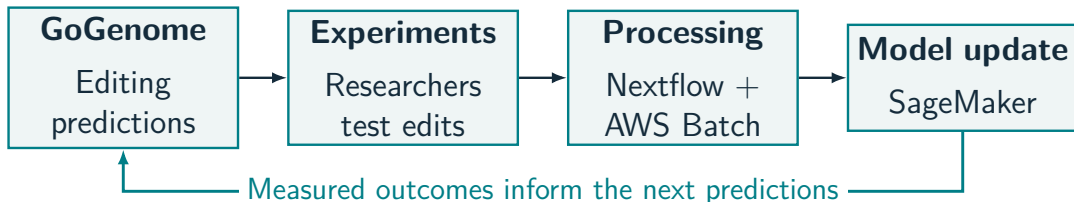
14,000+ datasets



Each output records how the analysis produced it.

Hitz et al., including Eliaz, [The ENCODE uniform analysis pipelines \(2023\)](#). Coauthored infrastructure.

CRISPR-IL fed measured outcomes into the next design round.



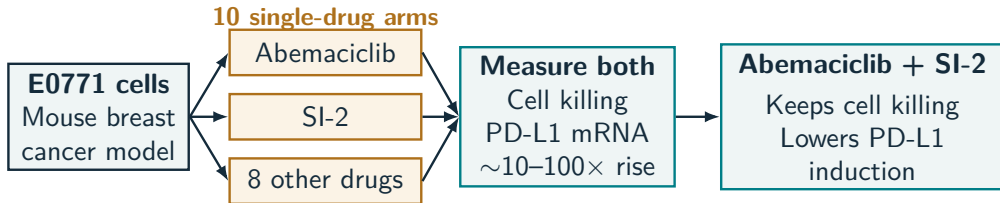
> 20 **TB** raw sequencing data in S3

GOLD: billions of indexed rows, reported feature lookup < 1 s

Noam Barkai & Yossi Eliaz, [A gene-editing prediction engine with iterative learning cycles built on AWS](#), 15 Feb 2022. Consortium work.

Ten drugs led to a pair with lower PD-L1.

10 drugs/candidates examined: 6 across 4 breast-cancer cell models; 4 more in E0771.



PD-L1 is a protein that can dampen immune attack. Compare both effects.

10 candidates $\longrightarrow (10 \times 9)/2 = 45$ **possible two-drug pairs**

45 possible pairs at one dose each. Selected-pair result: 72-hour assay in E0771 cells.

Parallel forks follow this pattern: fan-out trials, fan-in results.

Gilad, Eliaz et al., *Scientific Reports* 9:15099 (2019), Figs. 1, 3 and S2. Pair count is a combinatorial illustration.

Millions of cars build Mobileye's Roadbook.



Mobileye test vehicle, Manhattan, June 2021.

My work: junction perception, 2021–2024.

Pedestrian → brake → next camera frame.

34 billion road-miles in 2025 $\approx$ 1,080 road-miles/second (annual average).

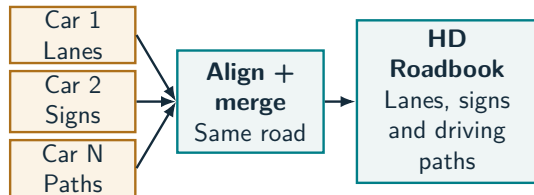
Parallel observations → align + merge → Roadbook → back to cars.

Software forks: failing test → try separate edits → compare checked results.

Mobileye, 21 July 2026; REM workflow. Photo: Mobileye / Intel press kit. Rate is derived from the 2025 total.

8M+ REM harvesting vehicles

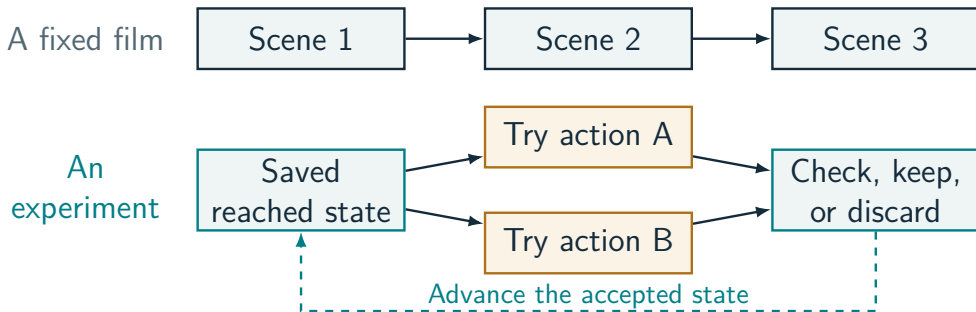
Collect road observations continuously, worldwide.



REM: Road Experience Management.

HD: high-definition road map.

Forkable computation lets us try different futures from the same state.



Organizing analogy and proposed execution model. Stored-program hardware remains the substrate.

The forkable API prepares once and runs alternatives in private children.

Try the same three edits, run the same checks, select a result.

Python loop

```
out = []
for edit in edits:
    env = prepare()
    env.apply(edit)
    out.append(test(env))
best = select(out)
```

3 prepared environments.

Proposed API sketch. Both columns are Python; each check returns its artifact and evidence.

Forkable API

```
parent = checkpoint(prepare())
children = fork(parent, n=3)
execute(children, edits)
out = evaluate(children, test)
best = select(out)
```

1 preparation; 3 private children.

1 MOTIVATION

Caching lets repeated experiments reuse their expensive build work.

Fresh compile  ~46 s

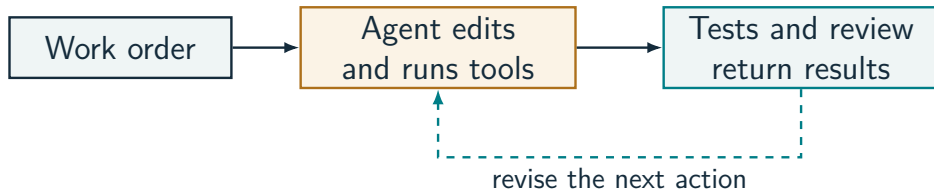
Restored cache  ~13 s

Hot cache  ~3 s

Tokio example: restoring the cache saved about 33 seconds per compile.

Eliaz, [Tokio issue #8200](#), June 2026: external Incredibuild demo. Artifact caching.

An agent changes the environment that executes its next experiment.



Airflow Factory: sandboxes through Docker Sandboxes (sbx); hosted models through Databricks.

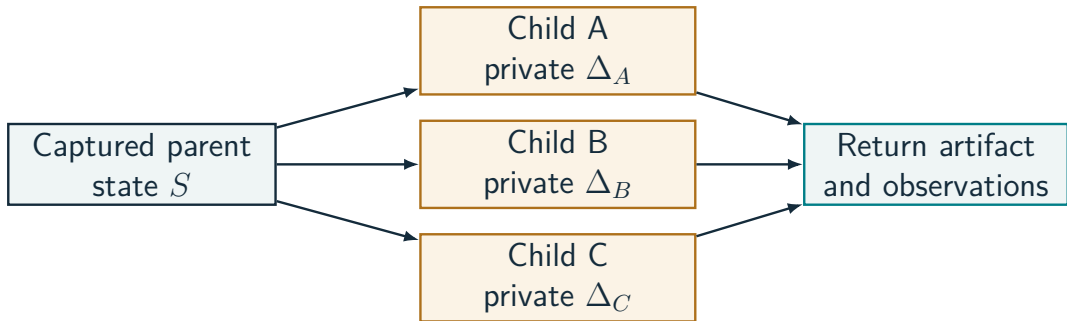
A sandbox is a policy boundary; VMs and containers are implementation choices.

Term	What it provides	Example
VM	Separate guest kernel	KVM + QEMU
microVM	Guest kernel; fewer devices	Firecracker
Container	Processes share a Linux kernel	Namespaces; cgroups
Sandbox	Policy over access and actions	Container or VM
cgroups	Resource usage and limits	CPU, memory, I/O

Docker Sandboxes (sbx): microVM + private Docker daemon.

Docker Sandboxes architecture; Firecracker; Linux cgroup v2.

A fork creates private continuations from one captured execution state.



Execution abstraction; backend-specific state and external-effect semantics.

Different fields use the same pattern: explore alternatives, then check the results.

The common pattern: shared setup → different trials → gather results.

Security and AI Changed inputs, model variants and candidate programs

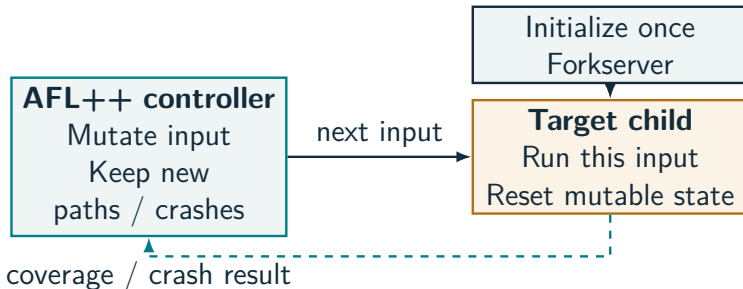
Genomics and physics Gene perturbations, drug partners and orbit connections

Our software factory Repairs, tests and delivery of the resulting patch

The examples show exploration and gathering; the runtime chapter turns the pattern into an execution contract.

AFL++ finds bugs by changing inputs and reusing the initialized program.

AFL++ is a fuzzer: it tests changed inputs and follows new code paths.



Persistent mode: about 1,000 inputs per child, then start a fresh child.

[AFL++; persistent-mode documentation.](#)

AFL++ reports 10–20 times faster execution in persistent mode.

10–20×

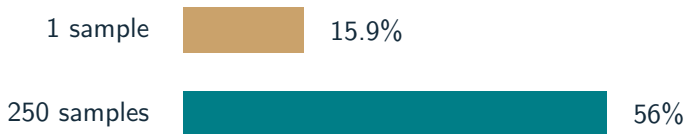
Typical speedup reported for persistent mode

One child amortizes process creation across many inputs.

Correct per-input reset keeps the reused state useful.

[AFL++ documentation](#); published typical gain, not our benchmark.

250 samples raised SWE-bench Lite coverage from 15.9% to 56%.

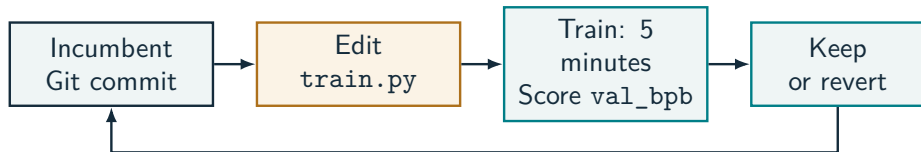


DeepSeek-Coder-V2-Instruct; pass at least once

Coverage means a correct patch exists; a selector must recognize it.

Brown et al., Large Language Monkeys (2024); SWE-bench Lite.

Autoresearch kept 23 experiments and lowered validation bits per byte.

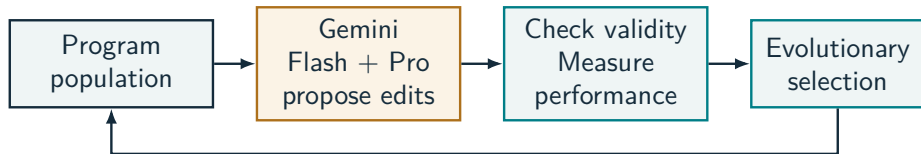


126 attempts: 23 kept, 102 discarded, 1 crash.

0.997900 $\rightarrow$ 0.969686: a 2.8% reduction; lower is better.

Karpathy, autoresearch protocol; reported session.

AlphaEvolve recovered 0.7% of Google's worldwide compute.

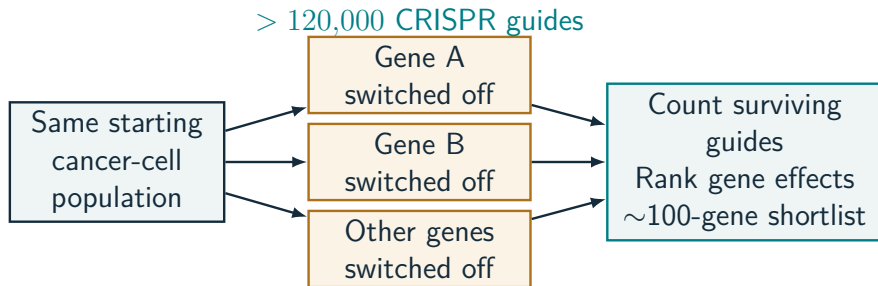


A deployed Borg scheduling heuristic recovered 0.7% on average.

Google DeepMind, AlphaEvolve (2025); production application.

A pooled gene-editing screen tested 19,050 genes to find drug partners.

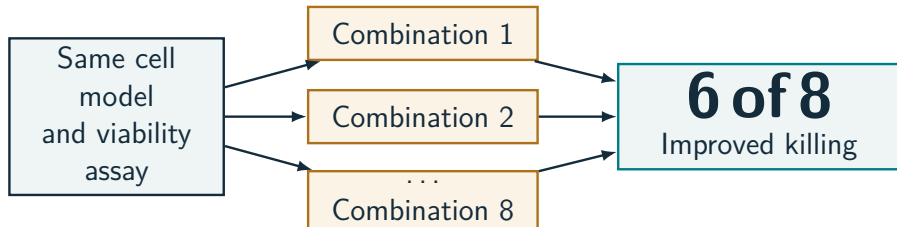
Perturbation: switch off a gene, then measure the cells' response.



Compare SI-12 with a control. SI-12 inhibits SRC-3, a protein that supports cancer-cell growth. Gilad, Eliaz et al. (2021); GeCKOv2 library; Terrace + DRACO analysis.

Six of eight drug combinations improved killing of breast cancer cells.

MCF-7: human breast cancer cells grown in the laboratory.

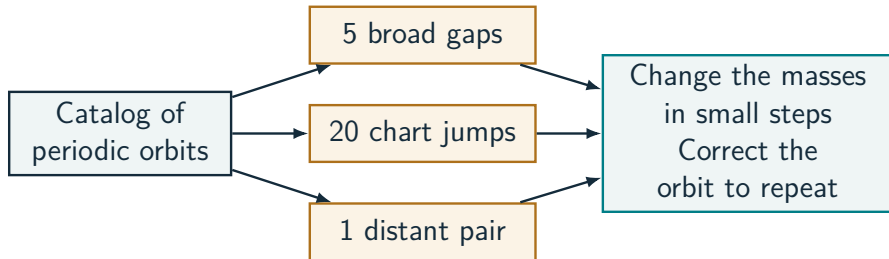


Fan-out: separate combinations. **Fan-in:** compare measured effects and retain promising candidates.

Gilad, Eliaz et al. (2021), Figs. 2–4. ≥ 4 technical repeats per point; ≥ 2 independent experiments per plot.

We tested whether apparently separate orbit families actually connect.

With **Ori Chamo**: 135,445 catalogued three-body orbits that repeat.



Numerical continuation: follow a family of solutions while changing a parameter.

Chamo–Eliaz preprint (2026); Three-Body Orbit Atlas.

All 26 selected connections survived checks in both directions.



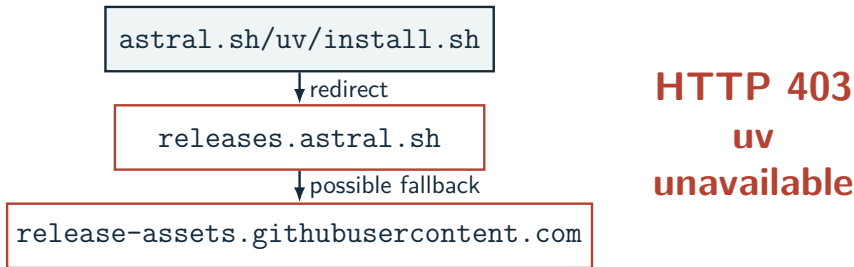
The selected links connect within **one sampled solution component**.

Representative stability changes were independently checked at **60-digit precision**.

Chamo–Eliaz preprint; Atlas numerical records.

The installer failed because its download destinations were blocked.

Task: let the sandbox install **uv**, the Python package manager.



Fix: allow the two destinations; align **six files** across policy, scripts, tests and docs.

Our Airflow Factory ran the repair serially in sandboxes (sbx); hosted model through Databricks.

Installing uv led to a 44-minute repair loop costing \$10.33.

Work	Time	Model calls
Specify and plan the change	3:41	\$0.96
Edit, test and repair	20:01	\$4.50
Review, repair and recheck	18:55	\$4.87

43:44 elapsed. Recorded stages: 42:38. Model calls: \$10.33.

A failed concurrency test redirected the work from installer policy into job dispatch.

Recorded session times include tool activity. Model costs use reported list prices.

Repairs and reviews consumed most of the recorded time and model cost.

Recorded time  59% / 25.2 min

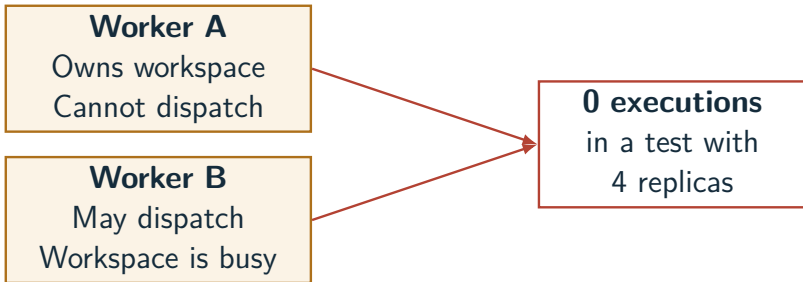
Model cost  75% / \$7.75

Two repairs + two reviews. Sandbox setup took **20.3 seconds**.

The next improvement must reduce repeated repair work or improve its checks.

One worker owned the workspace; another had permission to run.

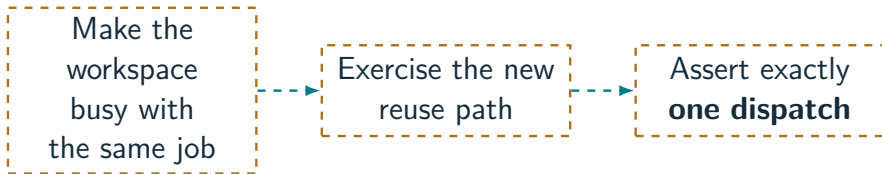
The repair agent's diagnosis: workspace ownership and dispatch permission split.



Repair: let the authorized worker reuse the active workspace for the same job.

The suite had zero failures, but the new recovery path still needed a test.

Reported suite: **1,972** cases, including skips; zero failures.



Review requested this targeted test; the final patch did not contain it.

Cleanup erased an approved patch before it was safely exported.

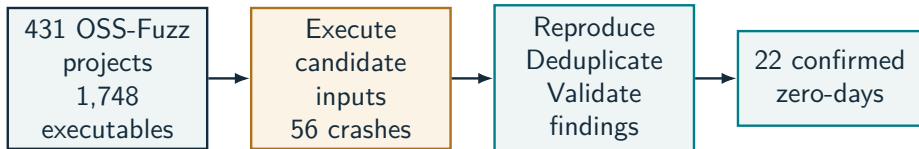
A second run: **about 26 minutes, \$3.43, no delivered pull request.**



Delivery found an extra review-archive file outside the approved file set.

Save the patch and test records outside the sandbox before cleanup.

CyberGym validated 22 zero-days after a 56-crash campaign.



OpenHands + GPT-5 proposed inputs; sanitizer runs produced candidate crashes.

[Berkeley RDI](#), [CyberGym](#); [CyberGym report](#); latest-code campaign.

A forkable runtime must preserve state, gather evidence and control acceptance.

The examples are complete. Now we turn their pattern into a system.

Run	What state is reused, and when is reuse cheaper?
Gather evidence	What ran, and how much information do its results add?
Accept	Which checked result may continue or be published?

The following chapters cover runtime costs and boundaries, information aggregation, then acceptance.

The next action determines which state a continuation needs.

Next action	Mechanism	State received
Edit source or build	Worktrees and build cache	Files and cached inputs
Continue live execution	Process or VM checkpoint	Supported memory and runtime state
Reuse an observation	Replay	Recorded execution or response

CRIU, gVisor, Firecracker and rr documentation; SnowFlock, EuroSys 2009; DeepSeek-V4 §5.2.5.

Our paper recorded 256 sandbox creates with a 3.44-second median.

Operation	Recorded result
Default sandbox environment	256 / 256 creates completed
Concurrency requested	8 calls at a time
Creation API latency	Median 3.44 s; p95 9.00 s

p95 means **95% of recorded calls completed within 9.00 seconds.**

[Eliaz, arXiv:2607.09689v4](#), Table 2. Creation API round trip; teardown excluded.

Isorun led the October 2 sandbox startup benchmark at a 73 ms median.

Time to interactive: create a sandbox and run its first command.

Provider	Median	p95	Success
Isorun	73 ms	78 ms	100 / 100
Miosa	163 ms	188 ms	100 / 100
Archil	238 ms	250 ms	100 / 100

ComputeSDK: 100 concurrent requests per provider, 2 October 2026.

Cheap starts raise the bar for state reuse.

[ComputeSDK Burst TTI](#) and [dated raw results](#). Different endpoint from our creation trace.

Reusing state pays off when it avoids more setup than it adds.

Saved repeated setup $>$ save once + restart children + private changes

$$(N - 1)P > H + N(R + D)$$

N Number of alternative runs

H Save the shared state once

P Setup each run would repeat

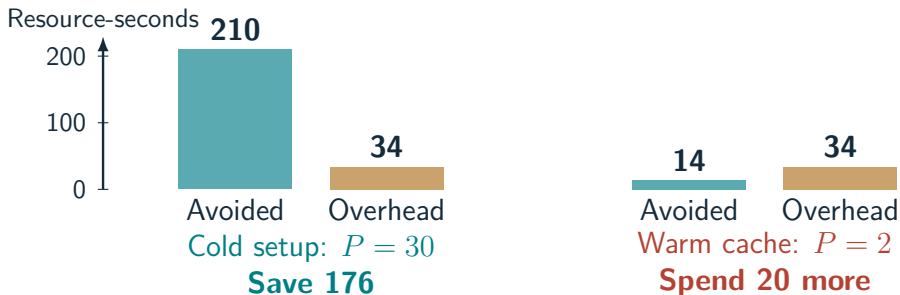
R Start a run from saved state

D Extra cost of private changes

Illustrative total-resource comparison. Common experiment work cancels; elapsed time also depends on scheduling.

Restoring wins against cold setup but loses against this warm cache.

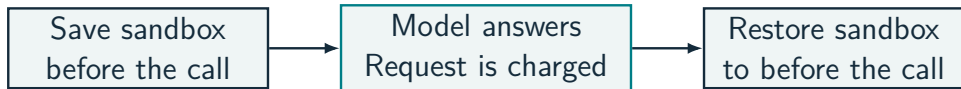
Eight alternative runs: $2 + 8(1 + 3) = 34$ resource-seconds of reuse overhead.



Illustrative additive resource model; $H = 2$, $R = 1$, $D = 3$ in both comparisons.

Restoring a sandbox does not undo an API call or its bill.

Example: a worker calls a hosted model after a checkpoint.

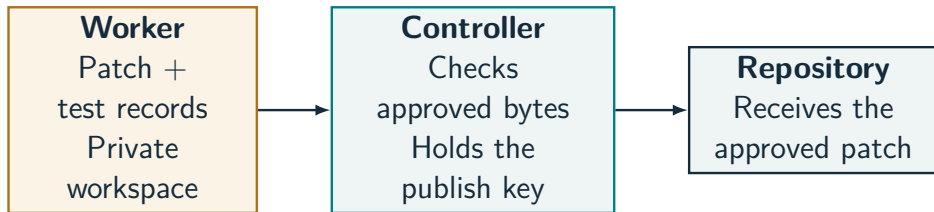


The external request has already happened.

Record the outcome outside the sandbox before deciding to retry.

Output-commit precedents: Speculator (2005), external synchrony (2006), Remus (2008).

Workers propose patches; the controller holds the key to publish them.

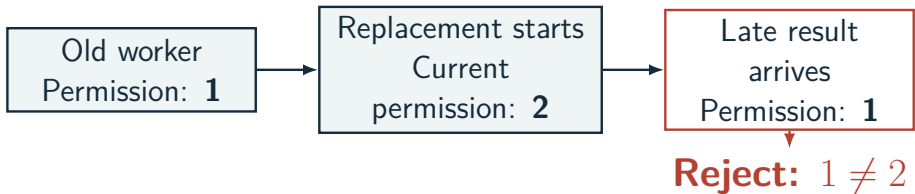


The worker can change its sandbox. Publication requires a separate decision.

Controller-held publication exercised in our factory; capability-security precedents: Capsicum and CHERI.

Once a worker is replaced, it can no longer publish results.

An **epoch** is the current version of a worker's permission.



The controller checks permission against its own current record.

Proposed publication protocol; leases and fencing in distributed systems.

Combining branch results should add information while accounting for overlap.

Fan-out explores alternatives. **Fan-in** turns their results into a useful decision.

Choose or merge artifacts: check the final code, model or candidate.

Pool measurements: use their reliability and shared information.

Goal: reduce uncertainty without counting the same information twice.

Next: merge rules, overlapping observations, statistical weights and the resulting precision.

Choosing a result, merging code and pooling measurements are different operations.

Operation	Example	What is checked?
Select	Pick one of three model variants	The chosen model on evaluation data
Compose	Merge two code changes	The actual merged code
Pool	Combine estimates of one effect	Noise and information they share

The output determines the merge rule.

Factory artifact checks and the numerical reducer illustrate different return paths.

Code changes that pass in pairs can fail when all three are merged.

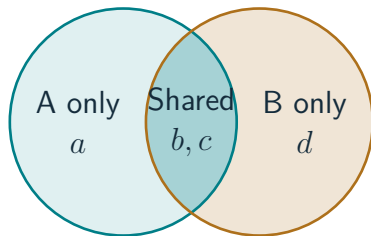
Example: each change starts a **1 GB worker**. The server has **2 GB**.

Merged changes	Memory used	Fits?
A, B or C alone	1 GB	✓ yes
A+B, A+C or B+C	2 GB	✓ yes
A+B+C	3 GB	✗ no

Run the checks on the final merged code before publishing it.

Illustrative resource-capacity example; successful pair tests do not cover every larger combination.

Set union keeps shared observations once; mutual information describes dependence.



**Union: 4 distinct
observations**

$$A \cup B = \{a, b, c, d\}$$

**Intersection: 2
shared observations**

$$A \cap B = \{b, c\}$$

Entropy H : uncertainty. Mutual information I : dependence.
For random variables: $H(X, Y) = H(X) + H(Y) - I(X; Y)$.

Set diagram: exact illustrative observation IDs. Entropy identity: [MIT information-theory notes](#).

Our reducer catches duplicate observation IDs and keeps their source history.

Reducer: the code that combines returned numerical summaries.

Returned records

Reducer behavior

Observation 42 arrives twice

Reject the duplicate

Observations 42 and 43 arrive

Combine summaries; retain both IDs

Next step: use recorded shared sources to calibrate statistical weights.

Reference reducer: IDs and history are implemented; covariance-aware weights are the proposed extension.

For estimates of one quantity, covariance guides the pooling weights.

Combine estimates x_i using weights that sum to one: $\hat{\mu} = \sum_i w_i x_i$.

$$w = \frac{\Sigma^{-1} \mathbf{1}}{\mathbf{1}^T \Sigma^{-1} \mathbf{1}}$$

Σ : the error covariance matrix—how noisy estimates are and how their errors move together.

$\mathbf{1}$: a vector of ones. Independent errors: weights are proportional to $1/\sigma_i^2$.

Shared errors: use the full covariance matrix.

Unbiased estimates of a common target; known positive-definite covariance. [Valassi & Chierici \(2014\)](#), [correlated measurement combination](#).

A worker that exaggerates its precision can dominate a weighted average.

$$\hat{\mu} = \frac{\sum_i p_i x_i}{\sum_i p_i} \quad p_i: \text{precision, or inverse variance. Larger } p_i \text{ means more weight.}$$

Synthetic example: one 2,000-point worker targets 17 and inflates reported information per point 50×.



Reference demo: one synthetic stress case. The weighting rule needs trusted precision estimates.

With shared errors, 100 estimates can reduce noise like about nine independent ones.

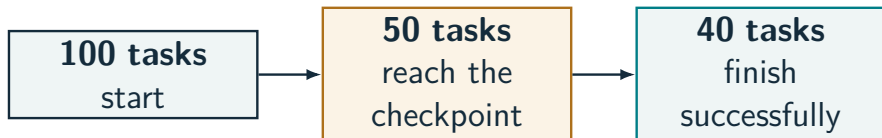
Example: 100 estimates with equal noise and pairwise correlation $\rho = 0.1$.

$$N_{\text{eff}} = \frac{N}{1 + (N - 1)\rho} = \frac{100}{1 + 99(0.1)} = 9.17$$

	Independent	Correlation 0.1
Noise relative to one estimate	0.10σ	0.33σ
Signal-to-noise gain for fixed signal	$10\times$	$3.03\times$

Exact mean-variance illustration: same marginal noise σ , common pairwise correlation, unchanged signal.

If 50 of 100 tasks reach a checkpoint and 40 finish, overall success is 40%.



From the checkpoint: $40/50 = 80\%$. From the start: $40/100 = 40\%$.

Both rates are useful. State which starting population you are measuring.

Illustrative counts: $P(\text{success}) = P(\text{checkpoint}) P(\text{success} \mid \text{checkpoint}) = 0.5 \times 0.8$.

A check record must say what ran, who ran it and which files it checked.

A **receipt** is a saved record of an operation and its result.

Label in our logs	What actually happened
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CI job passed	Evaluation was skipped; job ended in 3 seconds
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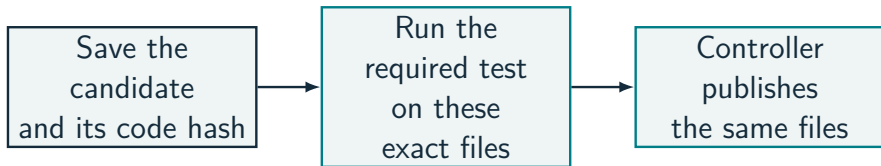
Human approved	Automation answered the approval prompt
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Code hash recorded	The helper code itself was unavailable
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Keep the actual check result and the exact code needed to reproduce it.

Three audit findings from our factory. A code hash identifies bytes; it does not store them.

The code we publish must be the same code that passed the required check.



Repair example: make the workspace busy, reuse it for the same job, and assert **one dispatch**.

If the files change after testing, check the new version before publishing.

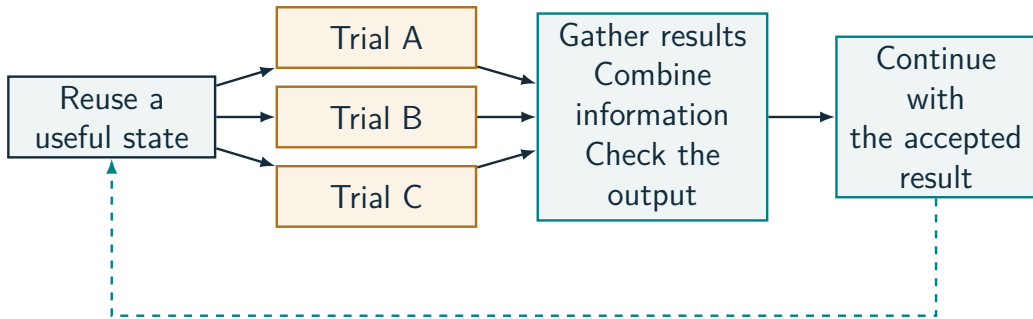
Proposed behavioral gate; builds on the factory's existing code-hash publication check.

Each use case needs a merge rule and a check matched to its output.

Use case	Gather / merge	Check before use
Model designs	Rank; select one	Evaluate the selected model
Code edits	Combine compatible edits	Test the merged code
Measurements	Remove duplicates; pool	Check noise and shared errors
Drug / orbit candidates	Rank or connect	Assays / numerical checks

Planned: restore versus warm reconstruction; siblings versus other families. Collection has not started.

Self-driving compute tries alternatives, combines what they teach us and continues.



Keep shared evidence once. Test the result. Let the controller accept it.

Forkable execution, information-aware aggregation and controller-held acceptance form the proposed system.